

BANDHU

AI Cognitive Care Companion

PROJECT NAVIGATION & DEVELOPMENT DOCUMENT

PND • Professional Technical & Product Roadmap

SIH26003

AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in
North Eastern Region (NER)

Ministry of Development of North Eastern Region (MDoNER)

Team: THE DEBUGGERS

Document version: 2.0 **Date:** September 2026

Purpose of this document

A single source of truth for what BANDHU is, what has already been implemented and verified, how the system is structured, what remains before the final demo checkpoint, and what the team plans to build if the solution is selected for further development.

Important: Implemented features and future roadmap items are deliberately separated throughout this document. Planned capabilities must not be presented as already deployed.

1. Document Map & Reading Guide

This PND is organized to let a judge, mentor, developer or future team member understand the project without needing to reconstruct the entire development history from Git commits.

Section	What it covers
2. Project Identity	Problem statement, organization, product identity, users and scope.
3. Problem & Solution	The user problem, design principles and BANDHU's response.
4. Product Scope	Current modules, completed features and explicit non-goals.
5. Architecture	Patient app, local persistence, adaptive engine, sync, backend and dashboard.
6. Patient Experience	Elderly-first UX, navigation, games, memories, routine and accessibility.
7. Caregiver Experience	Pairing, multi-patient dashboard, activity, progress, reminders and memory.
8. Context-Aware Help	Phase 1 screen-aware Help and Phase 3 live game-aware Help.
9. AI / ML Strategy	Tier-0 adaptation, research calibration and future consented personalization.
10. Data & Dataset Governance	Dataset principles, research data, licenses and no-synthetic-data rule.
11. Localization & NER	Language, voice, cultural memory and content-pack roadmap.
12. Privacy & Security	Current demo controls and post-selection hardening.
13. Implementation Record	Phase 1–3 status and current physical verification.
14. Repository Branding Cleanup	The supplied GitHub cleanup implementation report and compatibility rules.
15. Post-Selection Roadmap	What will be built if selected, in staged phases.
16. Testing & Release Gate	Final regression, APK, documentation and Git checkpoint.
17. Demo Story	Recommended end-to-end SIH software demonstration.
18. Sustainability & Risks	Open business-model decision and project risks.
19. Repository Documentation	README and PND linking guidance.

2. Project Identity, Scope & Boundaries

Field	Definition
Product	BANDHU — AI Cognitive Care Companion
SIH Problem Statement	SIH26003 — AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)
Organization	Ministry of Development of North Eastern Region (MDoNER)
Team	THE DEBUGGERS
Primary platform	Flutter patient application
Caregiver platform	React + TypeScript web dashboard
Backend	FastAPI REST service
Demo persistence	Drift / SQLite on the patient side, with demo backend support
Production persistence	PostgreSQL is planned if the project proceeds beyond demo stage
Core operating model	Offline-first with synchronization when connectivity is available
Product boundary	Assistive cognitive activities, memory assistance, routines, reminders and caregiver transparency — not diagnosis or treatment

Product promise

BANDHU helps an older adult engage with cognitive activities, remember personally meaningful information, follow daily routines and stay connected with a caregiver — while giving the caregiver a clear view of activity and engagement.

Non-clinical boundary

BANDHU is NOT a diagnostic system, medical predictor or clinical decision tool. Activity results are not clinical measurements. Medical concerns should be reviewed by qualified healthcare professionals.

3. Problem Understanding & Solution Strategy

The project is designed around a practical assistance problem: older adults can benefit from structured cognitive activities, familiar-memory prompts, routine support and reminders, while caregivers need visibility without constantly being physically present. In the North Eastern Region, language diversity, connectivity constraints and culturally familiar content make a one-size-fits-all digital experience insufficient.

BANDHU therefore combines five layers of assistance:

Layer	Purpose
Cognitive activity	Short, understandable exercises that encourage recall, attention, sequencing and pattern engagement.
Memory assistance	A personal memory bank, life-story foundation and reminiscence-oriented experiences.
Daily support	Routine recall, My Day, reminders and simple progress visibility.
Contextual assistance	A local Help assistant that understands the current screen and, in Phase 3, the active game state.
Caregiver transparency	Pairing and dashboard views for activity, progress, reminders, memories and synchronization.

Design principles

- **Offline first:** important interactions should not depend on continuous connectivity.
- **Elderly first:** readability, large touch targets, high contrast, calm navigation and reduced cognitive load.
- **Explainable:** deterministic assistance is preferred over opaque behavior in the demo.
- **Non-clinical:** no inference that converts gameplay into a medical conclusion.
- **Caregiver-transparent:** data should be understandable and patient-specific.
- **Region-aware:** language and cultural content should be validated rather than assumed.

4. Technical Architecture

BANDHU uses a layered architecture that keeps the patient experience functional locally while providing a clean path to a secure production backend.

Patient App (Flutter)

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Drift / SQLite Local Persistence

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Local Adaptive Engine + Context-Aware Help

↓

Offline Sync Queue / Sync Manager

↓

FastAPI REST API

↓

Production PostgreSQL (future)

↓

Analytics / Research ML (future expansion)

↓

Caregiver Dashboard (React + TypeScript)

Layer	Technology	Current position
Patient application	Flutter / Dart	Implemented and physically tested
Local persistence	Drift + SQLite	Implemented; offline-first foundation
Adaptive engine	Deterministic Dart service	Implemented
Context Help	Local HelpContext + game state	Implemented through Phase 3
Sync	Queue + FastAPI endpoints	Demo implementation
Backend	FastAPI	Demo implementation
Database	SQLite demo; PostgreSQL future	Production DB not required for current demo
Caregiver dashboard	React + TypeScript	Implemented
Research ML	Python / research datasets	Research calibration, non-diagnostic

Important demo architecture decision

The current demonstration does **not** require production PostgreSQL, Kubernetes, WebSockets or a large cloud deployment. Those are future engineering decisions that should only be introduced when the selected project has real deployment requirements.

5. Current Product — Implemented Scope

The following table describes the current software scope based on the development work completed before this PND.

Module	Current implementation status	Notes
Onboarding / login	Implemented	Patient onboarding, login foundation and Continue Offline.
Home	Implemented	Daily activity, next-activity guidance and Help.
Memory Match	Implemented + verified	Playable game, scoring, hints, live game context.
Pattern Recognition	Implemented + verified	Visual sequence activity with actual-state Help.
Daily Routine Recall	Implemented + verified	Routine sequencing and actual-state Help.
Reminders	Implemented	Routine, hydration and rest examples.
Progress	Implemented	Streak, completed activities and engagement chart.
Personal Memory Bank	Implemented locally	Drift/SQLite local persistence.
My Life Story	Functional	Local life-story foundation.
Familiar World / NER memory	Functional demo	Sample culturally familiar content foundation.
Family Connect	Demo/local foundation	Broader cloud sync is future work.
My Day	Functional	Routine timeline and self-reported daily reflection.
Mood Check-In	Functional	Self-reported only; no automatic inference.
Reminiscence Mode	Foundation	Text works; richer STT/voice remains future work.
Memory Through Music	Foundation	Playback demo; rights-cleared production content remains future work.
Talk to Me	Foundation	Deterministic local responses; controlled voice expansion future.
Caregiver pairing	Implemented + verified	4-digit code + QR + confirmation.
Multi-patient dashboard	Implemented	Patient selector and patient-specific views.
Phase 1 Help	Implemented + verified	Screen-aware local Help.
Phase 3 Help	Implemented + verified	Live game state, stuck detection, actual-state hints.

6. Elderly-First Patient Experience & Responsive Design

The patient application was designed for older users rather than treating the elderly experience as a smaller version of a conventional mobile app.

UX principle	Implementation direction
Touch targets	Minimum target size around 64×64 dp where practical.
Typography	Readable body text with 18sp+ targets in the elderly-first design system.
Contrast	High-contrast theme and accessible text hierarchy.
Visual language	Calm slate/navy, restorative sage and warm cream design direction.
Navigation	Simple bottom navigation on phones; NavigationRail/sidebar on larger layouts.
Responsive breakpoints	<600 phone; 600–1023 tablet; ≥1024 desktop.
Offline status	Clear local/offline state rather than hiding connectivity limitations.
Help	Non-intrusive Help button and gentle stuck prompt.
Accessibility	Voice-guidance foundation, high contrast, readable controls and future speech expansion.

Responsive strategy

The same product should remain usable on a phone, tablet and laptop. The layout changes by screen size rather than merely stretching the phone UI across a large display. The patient app keeps bottom navigation on phones and transitions to larger navigation structures on tablet/desktop layouts.

7. Cognitive Games & Adaptive Difficulty

The demo contains three cognitive activity experiences. They are presented as exercises and engagement activities, not clinical tests.

Game	Current behavior	Adaptive / Help state
Memory Match	Cards, matching, score, mistakes, flips and hints.	Board state, phase, level, matched pairs, flipped state and hint availability.
Pattern Recognition	Visual sequence with answer options and progression.	Pattern type, active sequence and state-aware guidance.
Daily Routine Recall	Routine steps are recalled/placed in chronological order.	Routine domain, current step and state-aware guidance.

Tier-0 deterministic adaptive rules

- **≥85% with low mistakes / fast response:** increase difficulty where the game supports it.
- **60–84%:** maintain current difficulty.
- **<60% or repeated failure:** reduce/support difficulty.
- Support can include longer display time, fewer items or additional instructions.
- The engine is deterministic and testable; it does not make a dementia judgement.

Critical safety rule

A hint must be derived from the actual current game state. The system must never invent a card, pattern element, routine step or answer that does not exist.

8. Context-Aware Help — Phase 1 and Phase 3

BANDHU Help evolved from a screen-aware assistant into a game-aware assistant. It remains local/offline and does not require an external AI API.

Phase 1 — Screen-aware Help

- Reusable HelpContext model.
- Current-route/screen detection.
- HelpContextService and route observation.
- Floating BANDHU Help button.
- LocalHelpEngine with deterministic, context-specific responses.
- Future AI-provider interface without requiring an external AI service in the demo.

Phase 3 — Game-aware Help

State captured	Why it matters
Current game	Lets Help speak about the actual activity.
Game phase	Prevents generic advice when the user is at a specific stage.
Difficulty	Explains difficulty without pretending it is a clinical score.
Score / progress	Provides relevant activity context.
Mistakes / hints used	Supports stuck detection and careful guidance.
Elapsed time	Used conservatively for inactivity detection.
Available hints	Lets Help expose only valid actions.
Board/pattern/routine payload	Enables actual-state-derived hints.

Stuck detection policy

The verified Phase 3 design uses a 20-second idle threshold, a mistake-based trigger, a 45-second cooldown and a short grace period. The prompt is deliberately gentle: **“Need a little help?”** with **Help me** and **Not now**. It should never flash aggressively or interrupt normal gameplay.

9. Caregiver Connection & Dashboard

Caregiver transparency is a central product feature. The dashboard is not intended to turn the caregiver into a clinician; it provides understandable activity and engagement information.

Pairing protocol

- 1 Caregiver creates a pairing request.
- 2 A random 4-digit code is generated.
- 3 A cryptographic pairing token is generated.
- 4 The QR payload uses the existing internal `smriti://pair?...` scheme for compatibility.
- 5 The pairing request is time-limited.
- 6 Patient confirms the pairing.
- 7 The caregiver can then select the connected patient.

Current pairing properties

Property	Current design
Code	4 digits, generated in the 1000–9999 range.
Token	Cryptographically generated token.
Expiry	10-minute pairing window in the implemented flow.
Status model	PENDING / USED / EXPIRED / CANCELLED.
QR	QR payload points to the existing internal pairing URI.
Patient confirmation	Required before connection completes.
Multi-patient	Caregiver can switch between connected patients.

Dashboard scope

- Overview and selected patient
- Activity history
- Memory
- Reminders
- Progress chart
- Sync status
- Patient switching
- Caregiver login/session foundation

10. Data, AI/ML & Personalization Strategy

The project uses a deliberately layered AI strategy. The current demo prioritizes deterministic behavior and explainability. Research ML is separated from product decisions that could be interpreted clinically.

Tier 0 — product adaptation

The current adaptive engine adjusts difficulty using observable activity signals such as accuracy, response time, mistake count, hints and streak. This is a rules-based interaction policy.

Tier 1 — research calibration

- Previously evaluated OpenNeuro Stroop data: 28 subjects in the referenced dataset.
- Previously evaluated OpenNeuro NYU Slow Flanker data: 26 healthy adults in the referenced dataset.
- Research work included 85 raw files, 54 participants and 4,585 empirical trials in the recorded project work.
- A Task-A random-forest benchmark reported MAE $\approx$ 126.45 ms, RMSE $\approx$ 170.85 ms and $R^2 \approx$ 0.2268.
- These are research engineering measurements, not clinical norms and not diagnostic thresholds.

Tier 2 — future consented telemetry

- Use only with explicit consent and appropriate privacy governance.
- Use telemetry for personalization, usability and activity adaptation.
- Do not convert telemetry into a dementia diagnosis or disease-severity score.
- Clinical research, if ever pursued, must be separately governed and validated.

AI/data rule

No synthetic data should be represented as real-world evidence. Dataset licenses, research-only restrictions and provenance must be documented before production use.

11. Dataset & Research Governance

The project has explicitly adopted a no-synthetic-data policy for evidence claims. Where datasets are used, their source, license/access status and intended use must be documented.

Resource / source class	Role in project	Governance note
OpenNeuro Stroop (ds000164)	Research calibration / engineering evaluation	Previously verified public dataset; respect dataset terms.
OpenNeuro NYU Slow Flanker (ds000102)	Research calibration / engineering evaluation	Previously verified public dataset; respect dataset terms.
Common Voice Assamese / Hindi	Speech/language resource evaluation	Use according to release/license terms.
AI4Bharat IndicConformer	Indian-language speech technology reference	Use according to project/license terms.
IndicVoices / IndicVoices-R	Indian speech/language research resource	Use according to release terms.
DementiaBank Pitt	Potential research-only resource	Controlled/research access; not a demo data source unless permitted.

Dataset selection rule

- Verify provenance before inclusion.
- Record license and access restrictions.
- Do not distribute restricted datasets with the repository.
- Do not use a research dataset as if it were a clinical benchmark for the NER population.
- Separate healthy-adult research calibration from any future clinical validation.

12. Localization, Voice & NER Cultural Memory

BANDHU's language and cultural architecture is designed for the North Eastern Region without assuming that one language or cultural context represents the entire region.

Current localization foundation

- English
- Hindi
- Assamese

Extensible NER language roadmap

- Bengali
- Meitei
- Bodo
- Mizo
- Khasi
- Garo
- Additional languages based on validated content, community input and resource availability.

Voice roadmap

- Improve offline-friendly voice guidance.
- Evaluate Indian-language speech resources where licensing and quality permit.
- Use voice for accessibility and interaction—not dementia detection.
- Provide non-voice fallbacks when speech services are unavailable.

Familiar World / NER Cultural Memory roadmap

- Regionally relevant objects, routines, foods, festivals, places and everyday contexts.
- Content packs with documented source/provenance.
- Community or subject-matter review before production use.
- Offline packaging so content remains usable without continuous connectivity.

13. Privacy, Security & Production Hardening

The demo demonstrates the user flow. Production deployment requires a stricter security and governance layer.

Area	Current demo	If selected
Identity	Demo authentication/session foundation	Production identity, token lifecycle, recovery and expiry.
Authorization	Patient-specific dashboard flow	Server-side ownership and role checks.
Pairing	4-digit + QR + time-limited token	Abuse controls, audit events and stronger account binding.
Storage	Local SQLite / demo storage	Encryption at rest where appropriate + secure cloud database.
Transport	REST demo	TLS, secure headers and hardened API.
Consent	Product boundary documented	Explicit data-sharing and synchronization consent.
Deletion/export	Future	User-appropriate export/deletion workflows.
Monitoring	Demo-level	Production logs, monitoring, alerts and incident response.
Backups	Demo not production	Secure, tested backups and recovery procedures.

Security principle

Do not add cloud infrastructure simply because it is technically possible. Introduce production infrastructure when it is needed for the selected pilot, and then apply appropriate security controls from the beginning.

14. Implementation History — Phase 1 to Phase 3

Phase	Purpose	Result
Phase 1	Context-aware local Help assistant	Implemented, tested and physically verified.
Phase 2	Memory, Life Story, Familiar World, Family Connect, My Day, Mood, Reminiscence, Music, Talk to Me, personalization foundation and responsive UI	Implemented/functional foundation; some items intentionally remain demo-level.
Phase 3	Game-aware Help, live game context, conservative stuck detection and actual-state-derived hints	Implemented and physically verified.

Phase 1 verification

- 279/279 tests passed in the recorded Phase 1 verification.
- Flutter analyze clean.
- Physical Android Help flow verified offline.
- No backend/pairing/dashboard regression reported.

Phase 2 verification

- 295 tests passed, 0 failed.
- Flutter analyze clean.
- Flutter web build successful.
- Responsive phone/tablet/desktop architecture implemented.
- No commit/push made at that stage.

Phase 3 verification

- 311 Flutter tests passed, 0 failed.
- flutter analyze: no issues.
- flutter build web: successful.
- Memory Match, Pattern Recognition and Routine Recall game-aware Help verified on physical Android.
- Offline Help verified with network disabled.
- No remaining bugs reported in the verification report.

15. Phase 3 Final Physical Verification Record

The supplied implementation report records an end-to-end physical verification on the Android device used for the demo. The following values are taken directly from that report.

Check	Recorded result
flutter analyze	PASS — No issues found.
flutter test	311 passed / 0 failed.
flutter build web	PASS.
Memory Match	PASS — live state received; actual board context available to Help.
Pattern Recognition	PASS — actual visual sequence recognized for relevant guidance.
Daily Routine Recall	PASS — active routine domain and chronological guidance verified.
Actual-state hints	PASS — suggestions reflected actual placed cards, active sequences and board states.
Stuck detection	PASS — naturally triggered after 22 seconds of inactivity.
Not now	PASS — prompt dismissed without interrupting gameplay.
Offline Help	PASS — network disabled; local guidance continued; no external AI/API required.
Regression	PASS — Home, Games, games, Progress, Reminders, Settings, Pairing and Phase 2 features checked.
Files changed during verification	None.
Remaining bugs	None reported.

Verification limitation

This confirms the current software behavior for the tested demo environment. It does not establish clinical efficacy, production security certification or population-level validation.

16. GitHub & Public Branding Cleanup — Implementation Record

The following section incorporates the implementation report supplied with this request. It is important because the product was rebranded from SMRITI to BANDHU while internal compatibility identifiers intentionally remain unchanged.

User-review constraints recorded in the implementation plan

- **Internal identifiers preserved:** ``smriti.db``, ``smriti.sqlite``, ``smriti_db``, ``smriti_user``, Dart classes such as ``SmritiRepository`` and ``SmritiTheme``, package ``org.sih26003.smriti.patient_app``, pairing scheme ``smriti://pair``, and storage keys.
- **Git history preserved:** existing commits such as ``SMRITI: working SIH demo checkpoint`` are not rewritten.
- **No commit or push during cleanup:** changes remain for user review until explicitly approved.

Documentation filename renames

Old public filename	New filename
SMRITI_MASTER_SPEC.md	BANDHU_MASTER_SPEC.md
docs/SMRITI_Work_Summary_Report.html	docs/BANDHU_Work_Summary_Report.html
SMRITI_Work_Summary_Report.pdf	BANDHU_Work_Summary_Report.pdf

Root documentation updates

- **README.md:** public title, clinical boundary, repository layout and product references changed to BANDHU; official SIH problem statement and MDoNER attribution retained.
- **walkthrough.md:** public titles/headings/UI references changed to BANDHU; internal database references retained.
- **BANDHU_MASTER_SPEC.md:** product descriptions and public notices updated; internal technical identifiers retained.

17. GitHub Cleanup — File-by-File Public Documentation Rules

The supplied cleanup implementation plan lists the following public-facing files for branding updates. The intent is not to replace every occurrence of the string `smriti`; it is to separate public product branding from technical compatibility identifiers.

Path / file	Public update
docs/architecture/ARCHITECTURE.md	Title and architecture header to BANDHU; language and clinical boundary text updated.
docs/BACKEND_SYNC.md	Title and diagram header to BANDHU; `smriti_db` / `smriti_user` retained.
docs/LOCALIZATION.md	Title and product description to BANDHU; `smriti_pref_language` retained.
docs/PHASE_08_ANALYTICS.md	Title and non-diagnostic statement to BANDHU.
docs/VOICE_ARCHITECTURE.md	Title, metadata and pipeline to BANDHU; `SmritiRepository` retained.
docs/decisions/ADR-001-foundation.md	Project name in context paragraph to BANDHU.
docs/demo/MEMORY_MATCH_DEMO.md	Title, introduction and assistive notice to BANDHU.
docs/demo/PATTERN_RECOGNITION_DEMO.md	Title, exercise description, run instructions and notice to BANDHU.
docs/demo/PHONE_DEMO.md	Title, run instructions and mobile confirmation to BANDHU.
data/DATASET_MANIFEST.md	Title and cultural asset provenance statement to BANDHU.
scripts/demo/README.md	Title and preview-tool description to BANDHU.
scripts/generate_summary_pdf.py	Report title, generated branding and output filenames to BANDHU.
docs/BANDHU_Work_Summary_Report.html	Report title and content branding to BANDHU.
scripts/demo/start_phone_demo.ps1	Console banner and launcher messages to BANDHU.
scripts/demo/generate_qr.py	Console branding to BANDHU; pairing protocol unchanged.
scripts/development/run_all_tests.ps1	Verification-runner branding to BANDHU.
.env.example	Header comment to BANDHU; database templates retained.

18. Branding Audit & Compatibility Rules

The cleanup verification plan requires a case-insensitive search over tracked files for `smriti`. Every result must be classified before deciding whether it should change.

Category	Rule	Examples
A — Public-facing	Must use BANDHU.	README title, documentation titles, UI-facing descriptions, public reports, public banners.
B — Internal/technical	Preserve unless a future compatibility migration is deliberately planned.	`smriti.sqlite`, `SmritiTheme`, package ID, `smriti://pair`, storage keys, internal routes.
Historical Git history	Do not rewrite.	Existing commit message `SMRITI: working SIH demo checkpoint`.

Required verification commands

1. Case-insensitive tracked-file search for `smriti`.
2. Classify all remaining occurrences.
3. Verify zero remaining public-facing SMRITI references.
4. Verify all remaining occurrences are justified internal/technical identifiers.
5. Run `git diff --check`.
6. Run `git status --short`.
7. Confirm zero commits and zero pushes were made during cleanup.

This is a compatibility-safe branding migration, not a blind global find-and-replace.

19. If Selected — Detailed Post-Selection Roadmap

The following roadmap describes what the team can develop after selection. The order is intentional: stabilize the verified demo first, then harden the platform, then expand regional/accessibility capabilities, then conduct controlled research and pilots.

Phase A — Demo stabilization

- 1 Freeze the current verified baseline.
- 2 Run final patient + caregiver regression.
- 3 Fix only verified blocking bugs.
- 4 Finalize public documentation and BANDHU branding.
- 5 Generate the final demo APK/web build.

Phase B — Production backend & security

- 1 Introduce production PostgreSQL when required.
- 2 Harden FastAPI authentication and authorization.
- 3 Implement stronger caregiver/patient ownership enforcement.
- 4 Improve sync conflict resolution and retry behavior.
- 5 Add production monitoring, backups and incident response.
- 6 Introduce secure deployment and encrypted transport/storage.

Phase C — Regional accessibility

- 1 Expand validated NER language packs.
- 2 Improve offline voice guidance.
- 3 Add reviewed culturally familiar content packs.
- 4 Improve accessibility for vision, hearing and motor constraints.
- 5 Validate usability with representative elderly users and caregivers.

Phase D — Personalization research

- 1 Collect only consented telemetry where justified.
- 2 Evaluate adaptive difficulty and engagement personalization.
- 3 Measure usability and activity outcomes using appropriate research methods.
- 4 Keep personalization non-diagnostic unless a separate, properly governed clinical research program is created.

Phase E — Pilot & scale

- 1 Pilot with selected caregivers / health workers.
- 2 Measure reliability, offline synchronization, accessibility and adoption.
- 3 Iterate based on real feedback.
- 4 Scale infrastructure only after pilot requirements are established.

20. Sustainability / Business Model — Deliberately Open

The team has intentionally kept the business/sustainability model open rather than locking the demo into an unsupported commercial claim.

Possible directions to evaluate after selection

- Institutional or public-sector deployment for community-care programs.
- NGO / elder-care organization partnerships.
- Supported caregiver dashboards for managed deployments.
- Grant or public-health program funding for pilots.
- A carefully evaluated service model if it preserves accessibility and affordability.

Evaluation criteria

- Affordability
- Privacy and trust
- Offline access
- Maintenance and support
- Language/accessibility cost
- Fit with public/community-care deployment

Decision status: **OPEN**. This section is a planning option, not a finalized business model.

21. Risks, Constraints & Mitigation

Risk / constraint	Mitigation
Poor connectivity	Offline-first persistence, local Help and synchronization queue.
AI hallucination / incorrect hints	Actual-state-derived hints and deterministic local Help.
Medical overinterpretation	Strict non-clinical wording and no diagnostic models.
Privacy exposure	Minimal data collection and production security hardening before deployment.
Patient data mixing	Patient-scoped data and server-side authorization in production.
Language mismatch	Regional validation and extensible localization.
Cultural assumptions	Community review and provenance for cultural content.
Dataset restrictions	License/access verification and research-only separation.
Feature creep	Release gate and no unnecessary new features before demo freeze.
Overengineering	Pilot-first architecture and staged production investment.
Accessibility barriers	Large targets, contrast, readable UI, responsive design and voice roadmap.

22. Final Release Gate & Remaining Work

The project is no longer at the feature-building stage. The remaining work is release discipline and final integration confidence.

Work item	Position
Phase 1	DONE
Phase 2	DONE / foundation implemented
Phase 3	DONE + physically verified
Patient app regression	Final cross-component check
Caregiver dashboard regression	Final check
Branding cleanup	Public documentation cleanup / audit
README → PND link	To add and verify
Final APK/build	Generate after final regression
Final Git checkpoint	Only after all checks pass
Git push	Last step, after user approval
PPT / demo video	Handled by teammates; not part of this software workload

Recommended final sequence

- 1 Finish GitHub public-branding cleanup.
- 2 Run final patient + caregiver regression.
- 3 Verify README and PND are visible in the repository.
- 4 Generate final APK/build.
- 5 Review ``git diff --check`` and ``git status``.
- 6 Create the final BANDHU checkpoint commit.
- 7 Push only after the final review.

No new major feature should be started after this gate unless it fixes a verified blocking issue.

23. Recommended SIH Software Demonstration Story

The strongest demonstration is a single narrative showing the user problem, the assistive solution, the caregiver connection and the offline-first differentiator.

Step	Action	Message demonstrated
1	Launch BANDHU	Clear product identity + elderly-first interface.
2	Show Home / My Day	Daily assistance rather than a standalone game.
3	Open Memory Match	Core cognitive activity.
4	Pause to trigger Help	BANDHU understands the user is stuck.
5	Open actual hint	Help uses live board state instead of generic text.
6	Continue game	Assistance is non-intrusive.
7	Show Memories / Reminders	Memory and daily-routine support.
8	Pair caregiver by QR / code	Transparent caregiver connection.
9	Open dashboard	Activity and progress are visible.
10	Show offline Help	Core assistance works without network.
11	Close with roadmap	Explain what selection enables next.

24. README & PND Integration

The PND should become a permanent repository artifact, not a detached file. The README should give a concise overview and point readers to the PND for the complete plan.

Recommended README block

■ Full Project Plan

BANDHU's complete technical roadmap, completed implementation scope, architecture, safety boundaries, verification status and post-selection development plan are documented in **BANDHU_PND.pdf**.

→ Check out **BANDHU_PND.pdf** for the complete project plan.

Recommended repository placement

- Keep `BANDHU\_PND.pdf` in the repository root for easy discovery.
- Place the README block after the main project overview / architecture and before very low-level developer details, or near the documentation section.
- Use a relative link such as `[■ View the Complete BANDHU PND](./BANDHU\_PND.pdf)`.
- Keep the public README concise; the PND contains the detailed roadmap.

Branding rule

GitHub's current public-facing presentation should consistently say **BANDHU**. Historical commit messages and internal compatibility identifiers may retain SMRITI/smriti where deliberately preserved.

25. Final Project Position

BANDHU has progressed from an initial concept into a working, physically verified SIH software demonstration. The core patient experience, cognitive games, memory assistance, routines, reminders, contextual Help, game-aware Help, caregiver pairing, multi-patient dashboard foundations, responsive design and offline-first behavior are in place.

The current objective is not to keep adding features indefinitely. The objective is to protect the working baseline, complete the final cross-component regression, clean the public repository branding, attach this PND to the README, generate the final demo build and create a final Git checkpoint.

If selected, the project can then move into production hardening, security and privacy maturity, NER language and cultural expansion, accessibility, consented personalization research and controlled pilot deployment.

Core principle: BANDHU should assist, personalize and connect — without turning gameplay, voice interaction or AI responses into a medical diagnosis.

END OF BANDHU — PROJECT NAVIGATION & DEVELOPMENT DOCUMENT